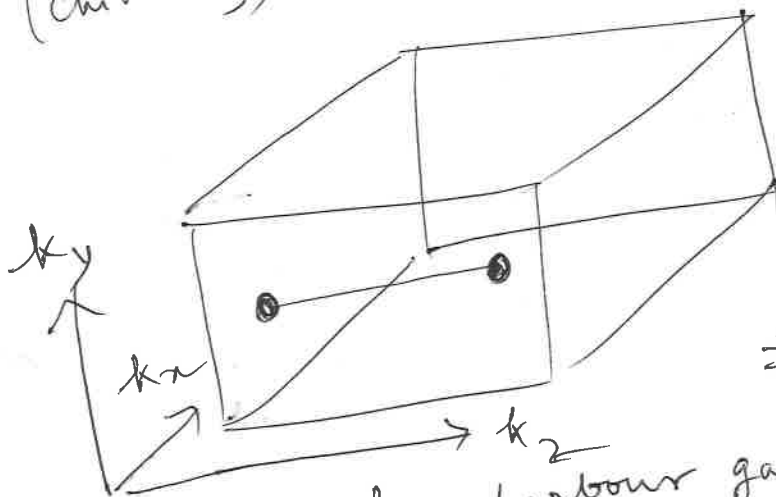


By stacking Chern insulators along the  $z$ -dim, employing a periodic B.C. along the  $z$ -dim, we get a  $k_z$ -dep. mass term vanishing at the 2 conjugate Weyl points [viz.,  $(-k_z^2 - k_0^2)\sigma_z$ ]. Thus, we get a model of a pair of Weyl points  $\Rightarrow W_+$  &  $W_-$  separated along  $k_z$  ( $\chi = \pm$  (chirality))

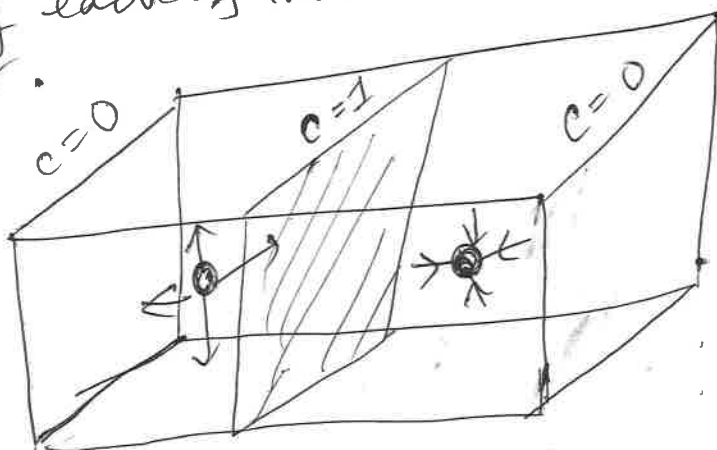


$$M(k_z) = k_z^2 - k_0^2$$

$$\Rightarrow M(k_z) = 0 \text{ at } k_z = \pm k_0$$

All 2D slices  $\perp k_z$  harbour gapped bands except at  $k_z = \pm k_0$ . In the region between the planes  $k_z = \pm k_0$ , i.e., for  $-k_0 < k_z < k_0$ ,

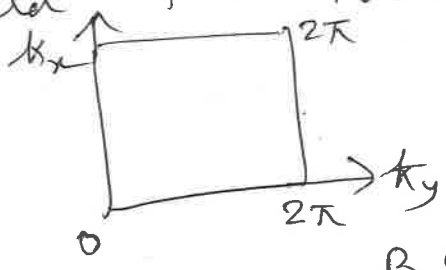
all  $k_x k_y$  planes must have Berry curvature flux lines penetrating them, endowing each of them with a nonzero Chern number.



see fig 3(a)  
of  
Reviews of Modern  
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[Lv, Qian & Ding]

A 2D gapped bandstructure in bulk gives rise to chiral edge states when it has a nonzero Chern number.

Suppose we start with such a system in the periodic situation comprising the  $k_x k_y$ -plane. Now unfold the Toric BZ as a rectangle



Impose open BC along ~~the~~  $x$ -dir<sup>n</sup>.  
 $\Rightarrow$  we get a chiral edge state (  $k_y$  is a good qm. number )  
 Bulk states  
 edge state along  $x$ -dir<sup>n</sup>

Each of the 2D planes will have an edge state if we take a slab of Weyl semimetal having surfaces  $\perp$  to  $x$ -axis, when we consider  $-k_0 < k_x < k_0$



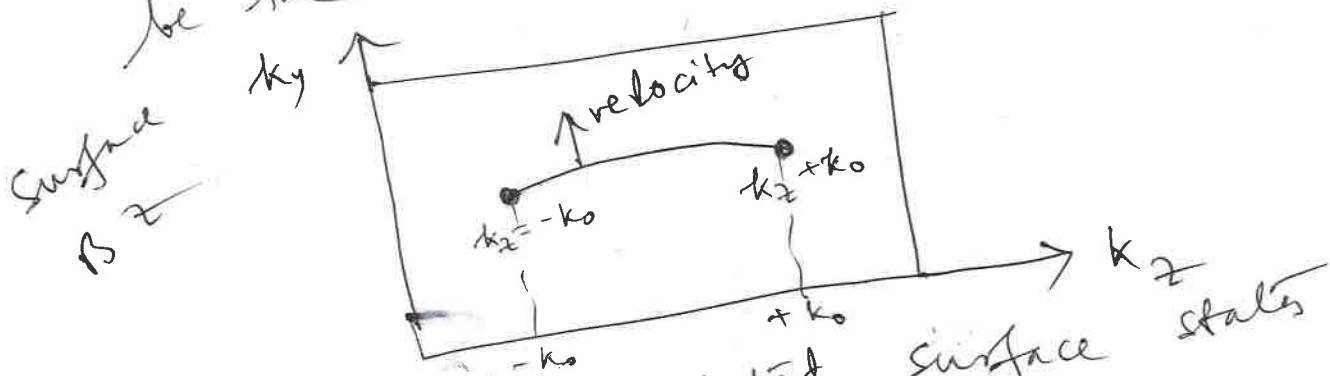
Construct surfaces as follows :-

(3)

$M(k_z)$ 
 $\xrightarrow[\text{to } \alpha\text{-dependent variable}]{\text{promote}}$ 
 ~~$M(k_z)$~~ 
 $\rightarrow \tilde{M}(k_z, \alpha)$

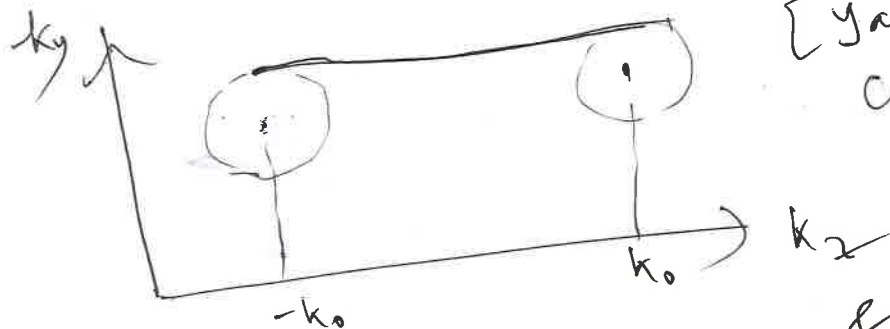
where  ~~$M(k_z)$~~  =  $\tilde{M}(k_z, \alpha) = \begin{cases} 0 & \text{for } \alpha > L \\ \text{ ~~$M(k_z)$~~  & for } 0 \leq \alpha \leq L \\ 0 & \text{for } \alpha < 0 \end{cases}$

when  $\mu$  or  $E_F = 0$ , we pick up states at energy 0 (i.e., at the energy of Weyl nodes, taken to be the zero of energy).



Energy of populated surface states are then at  $E_F = 0$

when  $\mu > 0$



See Annual Review  
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[Yan &  
Claudia Felder]

&  
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